Adaptive Gradient Harmonization: Mitigating Modality Dominance

Unified Representation Learning via Dynamic Convergence Balancing

Scholar Name: Lakshya
Registration No.: 25020010001

Supervisor: Prof (Dr.) Sandeep Joshi, Department of CSE
Faculty of Science, Technology and Architecture
School of CSE, Department of CSE
Manipal University Jaipur, Rajasthan, India

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Introduction

- **Multimodal Representation Learning**: The field aims to creating unified embeddings that integrate information from diverse sources: Text, Vision, Audio, and Sensors (IMU, Depth, Thermal).
- **Current State**: Recent "Unified Models" (e.g., ImageBind, Unified-IO 2) can theoretically handle arbitrary input combinations.
- **The Critical Challenge**: **Modal Dominance** (or "Modality Laziness").
- Models tend to over-rely on strong signals (RGB Images) and ignore weaker, noisier signals (Audio/Sensors), leading to poor robustness when the dominant modality is missing.

Problem Statement

This research addresses the **misalignment of convergence rates** and **gradient dominance** in unified multimodal learning.

- **Modality Laziness**: High-SNR modalities (Vision) dominate the joint optimization, suppressing learning from weaker ones.
- **Lack of Robustness**: "Unified" representations fail to generalize when key sensors fail or dominant modalities are corrupted.
- **Static Optimization**: Current methods use manual/static loss weighting, which is brittle and fails to adapt to training dynamics.

Literature Review (2023-2024)

Table: Summary of Foundational and State-of-the-Art Multimodal Models

Paper	Core Concept	Key Contribution	Identified Gap
Foundational Papers (BACD Framework)			
Gradient Blending (Wang et al., CVPR 2020)	Overfitting-based gradient weighting for multimodal fusion.	First systematic analysis: multimodal networks underperform due to dominant modality overfitting. 400+ citations.	Static/periodic updates; not real-time adaptive.
Greedy Nature (Wu et al., ICML 2022)	Coined "Modality Laziness"; proved it hurts generalization.	Conditional Learning Speed (CLS) metric to quantify and balance modality learning rates. 181 citations.	CLS metric is complex to compute; requires extensive calibration.
OGM-GE (Peng et al., CVPR 2022 Oral)	On-the-fly gradient modulation with generalization enhancement.	Dynamic noise injection to slow dominant modality. Proven at CVPR Oral (top 5%). 100+ citations.	Noise injection is a "black box"; lacks interpretability.
Cross-Modality Harmonization (Google, NeurIPS 2022)	Gradient conflict resolution at billion-sample scale.	Gradient realignment and curriculum learning based on conflict signals. Deployed at Google scale.	Designed for massive scale; heavy computational cost for smaller benchmarks.
Recent SOTA Models (2023-2024)			
ImageBind (Girdhar et al., CVPR 2023)	Joint embedding across 6 modalities (Image, Text, Audio, Depth, Thermal, IMU).	Zero-shot alignment using Images as the "binding" glue. No need for $O(N^2)$ paired data.	Heavily reliant on paired image data stability. prone to visual bias.
Unified-IO 2 (Lu et al., 2023)	Autoregressive transformer for generation & understanding.	Massive scale (7B params), handles diverse tasks (GRIT SOTA).	High computational cost; does not explicitly solve gradient suppression of weak signals.
Meta-Transformer (Zhang et al., 2023)	Frozen encoder framework for 12 modalities.	Decouples perception from semantic understanding using a shared token space.	Relies on frozen backbones; limited adaptation to novel, noisy sensor noise profiles.

Research Objectives

- **To Develop**: An **Adaptive Gradient Harmonization** algorithm ("Modality Fairness Controller") that dynamically re-weights loss components based on real-time generalization rates.
- **To Implement**: A **Dynamic Convergence Balancing** mechanism ensuring weak modalities (Audio/Sensors) reach optimal representation power alongside strong ones (Vision).
- **To Evaluate**: Rigorous testing on standard benchmarks (e.g., AudioSet, COCO) and robustness scenarios (inference with missing dominant modalities).

Proposed Methodology: The "Modality Fairness Controller"

Core Concept:

- Replaces static weights with dynamic scalars $\lambda_m(t)$.
- Monitors **Overfitting-to-Generalization Ratio (OGR)**.

Mechanism:

- **Throttle**: If Vision learns too fast ($\Delta L_v \gg$), reduce λ_{vision}.
- **Amplify**: If Audio lags, increase λ_{audio} to force feature extraction.

Proposed Methodology: System Architecture

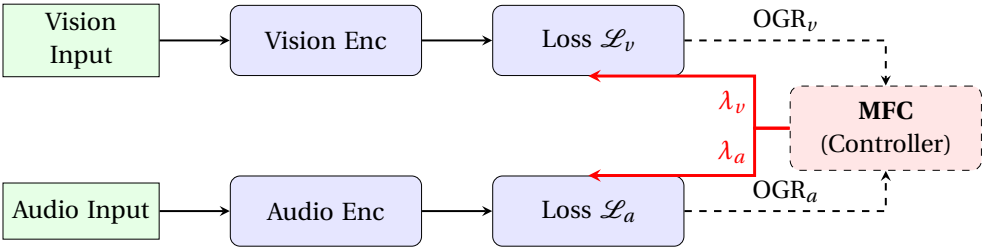


Figure: Adaptive Training Loop: The MFC monitors OGR and dynamically scales loss weights (λ) to balance convergence.

Key Contributions

- **Taxonomy**: A structured classification of "Modality Laziness" phenomena in current unified models.
- **Algorithm**: The **Adaptive Gradient Harmonization** method, a parameter-free approach to balancing heterogeneous sensor fusion.
- **Framework**: A Unified Transformer architecture integrated with the fairness controller, capable of robust performance even with missing data.
- **Benchmark**: Evaluation on modern multimodal datasets comparisons against static weighting baselines (ImageBind, Meta-Transformer).

Status Update

- **Phase 1: Literature Survey (Completed)**
  - Reviewed key papers: ImageBind, Unified-IO 2, Meta-Transformer.
  - Analyzed gaps in "Missing Modality" research (Wu et al. 2024).
- **Phase 2: Problem Formulation (Completed)**
  - Defined "Modality Laziness" as the core optimization bottleneck.
  - Formulated the hypothesis for Dynamic Convergence Balancing.
- **Phase 3: Resource Collection (Ongoing)**
  - Acquired datasets (AudioSet, COCO) and baseline codebases.

Schedule (Next 6 Months)

Month	Activity	Outcome
Month 7-8	Implementation of Baseline	Functional Unified Transformer
Month 9-10	Develop "Fairness Controller"	Working Adaptive Algorithm
Month 11-12	Evaluation & Thesis Writing	Comparative Results & First Draft

Figure: Project Timeline

Conclusion

- Current Multimodal models suffer from **Modality Laziness**, where strong visual signals suppress learning from audio/sensors.
- We propose a novel **Adaptive Gradient Harmonization** approach to dynamically balance learning rates across modalities.
- By forcing the model to learn "fairly", we aim to achieve **true unified robustness**, ensuring performance holds even when dominant sensors fail.

Selected References I


Wang, W., et al. "What Makes Training Multi-Modal Classification Networks Hard?" *CVPR*, 2020. (400+ citations)


Wu, Y., et al. "Characterizing and Overcoming the Greedy Nature of Learning in Multi-modal Deep Neural Networks." *ICML*, 2022. (181 citations)


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Thank You

Questions?